CHAPTER 8

ORGANIC CHEMISTRY: SOME BASIC PRINCIPLES AND TECHNIQUES

CHAPTER ESSENCE AT A GLANCE:
Covers carbon tetravalency, IUPAC organic nomenclature, isomerism types, electronic displacement effects, and reaction intermediate stabilities.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Carbon Hybridization & Bonding:** Carbon forms sp3 (tetrahedral), sp2 (trigonal planar), and sp (linear) hybrid orbitals governing organic geometry.

• **Inductive & Resonance Effects:** Inductive effect is sigma-bond electron shifting; resonance involves delocalized pi electrons over conjugated chains.

• **Hyperconjugation Stability:** No-bond resonance delocalization of sigma C-H electrons into empty p-orbitals stabilizes carbocations and alkenes.

• **Carbocation vs Carbanion Stability:** Carbocations are stabilized by +I and hyperconjugation (3° > 2° > 1°); carbanions are destabilized by alkyl groups (1° > 2° > 3°).

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\text{Carbocation Stability: } 3^\circ > 2^\circ > 1^\circ > CH_3^+$$

Carbocation Stability via Hyperconjugation

$$\text{Carbanion Stability: } CH_3^- > 1^\circ > 2^\circ > 3^\circ$$

Carbanion Stability Order

$$+I \text{ effect: } -C(CH_3)_3 > -CH(CH_3)_2 > -CH_2CH_3 > -CH_3$$

Electron Releasing Inductive Effect Hierarchy

$$\%N = \frac{1.4 \times M \times (V - V_1/2)}{W}$$

Kjeldahl Method for Percentage Nitrogen Estimation

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Inductive effect is distance-dependent and dies out beyond 3 carbon atoms; resonance delocalization spans the entire conjugated system.

• **Alert:** Do not confuse electrophiles (electron deficient, Lewis acids like BF3, AlCl3) with nucleophiles (electron rich, Lewis bases like NH3, OH-).

• **Alert:** In Lassaigne's test, Prussian blue colour confirms nitrogen via formation of ferric ferrocyanide: Fe4[Fe(CN)6]3.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Differentiate between homolytic fission and heterolytic fission of a covalent bond with suitable examples. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Homolytic Fission:** Symmetrical cleavage where each bonded atom takes one of the shared electrons, resulting in the formation of highly reactive, neutral free radicals (e.g., Cl-Cl -> 2 Cl· in sunlight). **[1 Mark]**

• **Heterolytic Fission:** Unsymmetrical cleavage where the more electronegative atom retains both shared bonding electrons, producing ions: carbocations and carbanions (e.g., (CH3)3C-Br -> (CH3)3C+ + Br-). **[1 Mark]**

• **Key Exam Result:** Homolysis generates free radicals; Heterolysis generates carbocations and carbanions.

2. Write the IUPAC names of the following organic compounds: (a) CH3-CH(CH3)-CH2-CHO, (b) CH3-CH=CH-COOH, (c) CH3-C(CH3)2-CH2-OH. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **(a) CH3-CH(CH3)-CH2-CHO:** Principal functional group is aldehyde (-CHO at C1). 4-carbon parent chain with methyl at C3. IUPAC Name: 3-methylbutanal. **[0.5 Mark]**

• **(b) CH3-CH=CH-COOH:** Principal group is carboxylic acid (-COOH at C1). 4-carbon chain with double bond at C2. IUPAC Name: But-2-enoic acid. **[1 Mark]**

• **(c) CH3-C(CH3)2-CH2-OH:** Alcohol (-OH at C1). 3-carbon parent chain with two methyl groups at C2. IUPAC Name: 2,2-dimethylpropan-1-ol. **[0.5 Mark]**

• **Key Exam Result:** 3-methylbutanal, But-2-enoic acid, 2,2-dimethylpropan-1-ol.

3. Explain the Inductive Effect with an example. Differentiate between +I and -I groups with two examples of each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** Permanent polarization of a sigma covalent bond caused by electronegativity differences between bonded atoms, transmitted along a carbon chain. **[1 Mark]**

• **+I Effect (Electron Donating):** Alkyl groups push electron density away from themselves into the carbon chain (+I groups: -CH3, -CH2CH3, -CH(CH3)2). **[1 Mark]**

• **-I Effect (Electron Withdrawing):** Strongly electronegative groups pull electron density towards themselves (-I groups: -NO2, -CN, -COOH, -F, -Cl). **[1 Mark]**

• **Key Exam Result:** Inductive effect is a distance-dependent sigma-bond polarization fading rapidly beyond 3 carbons.

4. What is Hyperconjugation? Explain why a tertiary carbocation is more stable than a primary carbocation on the basis of hyperconjugation. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** Delocalization of sigma electrons of a C-H bond of an alkyl group directly attached to an atom with an unshared p-orbital (or double bond) into the vacant p-orbital, also called no-bond resonance. **[1 Mark]**

• **Number of Hyperconjugative Structures:** Stability increases with the number of alpha-hydrogen atoms. Tertiary butyl cation (CH3)3C+ has 9 alpha-hydrogens and forms 9 hyperconjugative canonical structures. **[1 Mark]**

• **Comparison with Primary:** Ethyl carbocation CH3-CH2+ has only 3 alpha-hydrogens (3 structures). Methyl cation CH3+ has 0. Hence stability order: 3° (9 alpha-H) > 2° (6 alpha-H) > 1° (3 alpha-H). **[1 Mark]**

• **Key Exam Result:** Greater number of alpha-hydrogens in 3° carbocations maximizes hyperconjugative dispersal of positive charge.

5. Explain structural isomerism. Describe chain, position, functional, and metamerism with one example of each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Chain Isomerism:** Compounds with identical molecular formula but different carbon skeleton chains (e.g., n-butane and isobutane for C4H10). **[0.5 Mark]**

• **Position Isomerism:** Same carbon skeleton but different position of substituent or functional group (e.g., Propan-1-ol and Propan-2-ol for C3H8O). **[1 Mark]**

• **Functional Isomerism:** Identical molecular formula having different functional groups (e.g., Ethanol CH3CH2OH and Dimethyl ether CH3OCH3 for C2H6O). **[1 Mark]**

• **Metamerism:** Unequal distribution of alkyl groups on either side of the identical polyvalent functional group (e.g., Diethyl ether C2H5-O-C2H5 and Methyl propyl ether CH3-O-C3H7). **[0.5 Mark]**

• **Key Exam Result:** Structural isomers differ in atom connectivity and bonding order.

6. Define Electrophiles and Nucleophiles with two examples of each. Why does anhydrous AlCl3 act as an electrophile in organic reactions? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Electrophiles (Electron Loving):** Electron-deficient species that accept an electron pair (Lewis acids). Examples: H+, NO2+, Cl+, Carbocations, and neutral molecules like BF3, AlCl3. **[1 Mark]**

• **Nucleophiles (Nucleus Loving):** Electron-rich species carrying lone pairs or negative charge that donate electron pairs (Lewis bases). Examples: OH-, CN-, Cl-, H2O, NH3. **[1 Mark]**

• **Anhydrous AlCl3 Justification:** In AlCl3, aluminium has only 6 valence electrons around it, having an incomplete octet (electron-deficient). It acts as an electrophile by accepting an electron pair. **[1 Mark]**

• **Key Exam Result:** Electrophiles are electron pair acceptors; Nucleophiles are electron pair donors.

7. Describe the chemical principles of Lassaigne's test for detecting nitrogen, sulphur, and halogens in organic compounds. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Sodium Fusion Extract:** The organic compound is fused with metallic sodium to convert covalent nitrogen, sulphur, and halogens into water-soluble ionic sodium salts (NaCN, Na2S, NaX). **[1 Mark]**

• **Nitrogen Detection:** Extract boiled with FeSO4 and acidified with conc. H2SO4 produces Prussian blue precipitate of ferric ferrocyanide: Fe4[Fe(CN)6]3. **[1 Mark]**

• **Sulphur and Halogens:** Sulphur gives deep violet colour with sodium nitroprusside: [Fe(CN)5NOS](4-). Halogens acidified with HNO3 and AgNO3 yield AgCl (white), AgBr (pale yellow), or AgI (yellow). **[1 Mark]**

• **Key Exam Result:** Sodium fusion converts covalent heteroelements into easily testable ionic salts.